

- 3 Fig. 3.1 and Fig. 3.2 are diagrams of transverse sections of the human heart during different stages of the cardiac cycle. The sections pass through the heart at a level that is just above the valves.

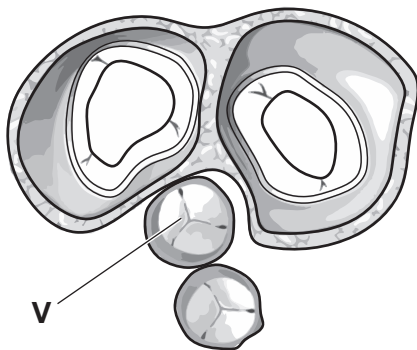


Fig. 3.1

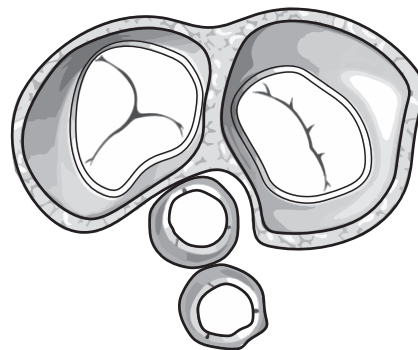


Fig. 3.2

- (a) (i) Name the valve labelled **V** on Fig. 3.1.

..... [1]

- (ii) With reference to the chambers of the heart and the main blood vessels, describe the flow of blood through the heart that occurs when the transverse section of the heart appears as shown in Fig. 3.1.

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 [2]

- (b) Identify the stage of the cardiac cycle shown in Fig. 3.2.

Give a reason for your answer.

stage of cardiac cycle

reason

.....

.....

[2]



- (c) The rate and rhythm of the heartbeat are controlled by an area of specialised muscle tissue in the wall of the right atrium, called the sinoatrial node.

Describe the sequence of events that control contraction of the ventricles during the cardiac cycle.

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.....[3]

[Total: 8]